

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0211

Neosho 138 kV
July 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0211. The Interconnection Customer proposes a 256.8 MW solar photovoltaic facility interconnecting at the Neosho 138 kV substation in the Virginia area. The request was received on March 06, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 150 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,927,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Salem Electric Department / Neosho 138 KV L
Contingency: Neosho / Montello Substation 138 KV
Loading: 155.5% (Rating: 165 MVA)
DFAX: 61.7%
Pre-project loading: 59.3%
2. Facility: Hancock / Ash Street 138 KV
Contingency: Beardsley St / Lewistown No. 1 138 KV
Loading: 131.5% (Rating: 232 MVA)
DFAX: 76.3%
Pre-project loading: 46.9%
3. Facility: Montello / Ash Street 138 KV
Contingency: Salem Electric Department / Neosho 138 KV
Loading: 124.1% (Rating: 203 MVA)
DFAX: 61.7%
Pre-project loading: 46.0%
4. Facility: Montello / Neosho 138 KV
Contingency: Salem Electric Department / Neosho 138 KV
Loading: 122.5% (Rating: 206 MVA)
DFAX: 61.7%
Pre-project loading: 45.4%
5. Facility: Monel / Candelaria 138 KV
Contingency: Clover / Bear 345 KV
Loading: 110.5% (Rating: 379 MVA)
DFAX: 5.0%
Pre-project loading: 107.1%

6. Facility: Salem Electric Department / Repeto 138 KV
 Contingency: Montello / Neosho 138 KV
 Loading: 107.4% (Rating: 211 MVA)
 DFAX: 61.7%
 Pre-project loading: 32.4%
7. Facility: Lewistown No. 1 / Hondo 138 KV
 Contingency: Hancock / Ash St 138 KV
 Loading: 106.4% (Rating: 185 MVA)
 DFAX: 47.4%
 Pre-project loading: 40.6%
8. Facility: South Crk / Wythe Substation 138 KV
 Contingency: Lyondell / Orlando Cogen Lp Substation 138 KV
 Loading: 105.8% (Rating: 255 MVA)
 DFAX: 5.1%
 Pre-project loading: 100.6%
9. Facility: Neosho / Beardsley St 138 KV
 Contingency: Hancock / Ash Street 138 KV line
 Loading: 105.7% (Rating: 204 MVA)
 DFAX: 100.0%
 Pre-project loading: 20.0%
10. Facility: Louisa Generation Facility 345-138 kV XFMR
 Contingency: Fluvanna P.S / Louisa Generation Facility 345 KV
 Loading: 104.0% (Rating: 462 MVA)
 DFAX: 5.0%
 Pre-project loading: 101.2%
11. Facility: Lewistown No. 1 / Beardsley St 138 KV
 Contingency: Hancock / Ash Street 138 KV
 Loading: 102.4% (Rating: 212 MVA)
 DFAX: 47.4%
 Pre-project loading: 45.0%
12. Facility: N Huntington / Ensln 138 KV
 Contingency: Moorland / Walerga- Antelope 138 KV Line
 Loading: 101.6% (Rating: 517 MVA)
 DFAX: 7.3%
 Pre-project loading: 98.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Salem Electric Department / Neosho 138 KV L	Rebuild line with higher-capacity conductor	\$749,000	100% (\$749,000)
Hancock / Ash Street 138 KV	Rebuild line with higher-capacity conductor	\$2,925,000	64% (\$1,864,000)
Ash Street / Montello 138 KV	Rebuild line with higher-capacity conductor	\$2,034,000	100% (\$2,034,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montello / Neosho 138 KV line	Rebuild line with higher-capacity conductor	\$1,008,000	100% (\$1,008,000)
Monel / Candelaria 138 KV	Rebuild line with higher-capacity conductor	\$28,168,000	6% (\$1,720,000)
Salem Electric Department / Repeto 138 KV	Reconductor line	\$884,000	100% (\$884,000)
Lewistown No. 1 / Hondo 138 KV	Reconductor line	\$1,079,000	100% (\$1,079,000)
South Crk / Wythe Substation 138 KV	Reconductor line	\$3,408,000	7% (\$236,000)
Beardsley St / Neosho 138 KV line	Reconductor line	\$2,374,000	100% (\$2,374,000)
Louisa Generation Facility 345/138 kV Auto	Replace transformer with higher-rated unit	\$10,781,000	7% (\$775,000)
Lewistown No. 1 / Beardsley St 138 KV	Reconductor line	\$2,038,000	100% (\$2,038,000)
Enslen / North Huntington 138 KV Line	Reconductor line	\$4,421,000	4% (\$166,000)
	Total allocated to project		\$14,927,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.